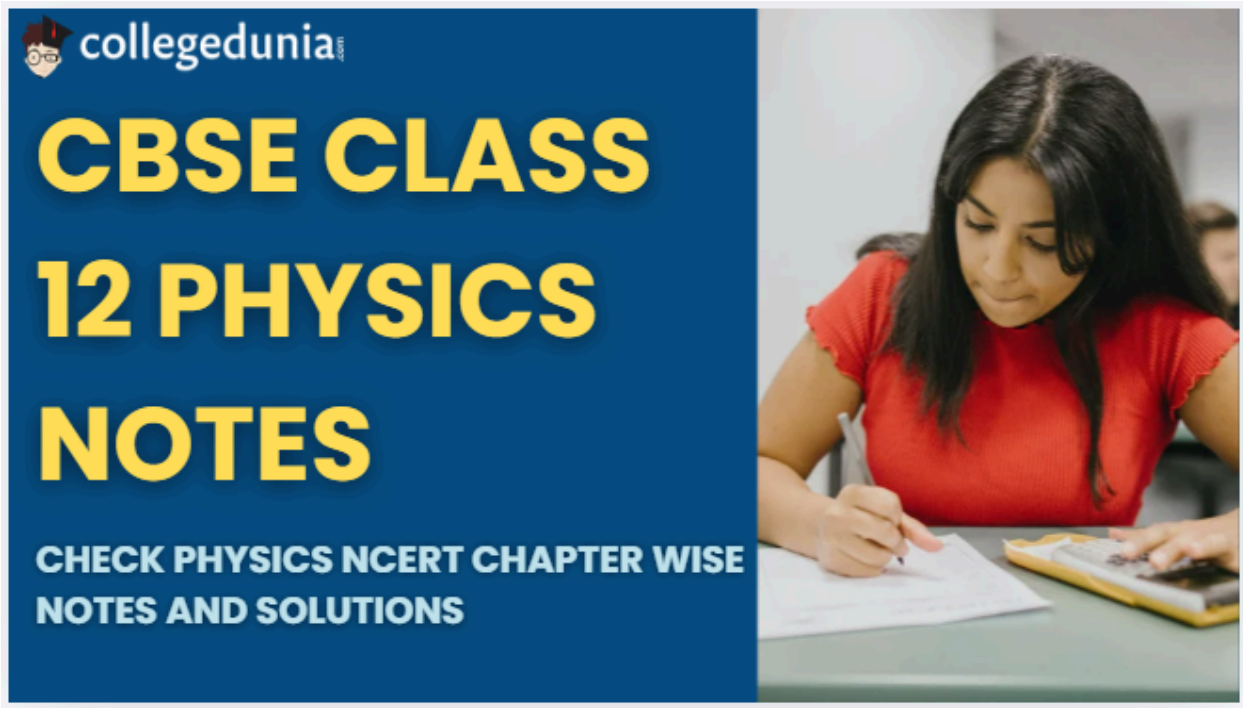
CBSE Class 12 Physics Notes: Check Physics NCERT Chapter Wise Notes and Solutions

CBSE Class 12 Physics notes, as per the latest CBSE Class 12 syllabus has been provided in this article. The **CBSE Class 12** notes consist of chapter-wise notes and NCERT Solutions to prepare for the CBSE Class 12 Board exam. Students can find links to NCERT Notes for all physics chapters of Class 12, organized by chapter number and title.

- There are a total of 14 chapters in the **CBSE Class 12 Physics syllabus**.
- **One chapter, Communication Systems (Chapter 15), has been removed from the CBSE Class 12 Physics syllabus and the NCERT textbook**
- These notes will be very useful during the preparation of the board exam.
- Class 12 Physics Notes will help in quick revision and provide an overview of the chapter.
- These notes cover all key concepts, theories, and formulas presented in a clear and simplified format.



Also Check:

- **NCERT Books Class 12 (Available): Download Free PDF for Class 12 Books Here**
- **CBSE Class 12 Chapter-Wise Weightage: Check Weightage and Important Topics**
- **CBSE Class 12 Question Paper 2026 with Solutions (Available)- Download Pdf**

- **CBSE 12th Exam Analysis 2026: Check Expected Difficulty Level, Subject-wise Weightage, and Question Papers PDF**

Chapter-Wise CBSE Class 12 Physics Notes

In this section, we have provided the Class 12 Physics Notes for important topics along with chapter-wise NCERT solutions.

Class 12 Physics Notes for Chapter 1: Electric Charges and Fields

The Class 12 Physics chapter 1 deals with the studies on **Electric Charges and Fields**. In this chapter the **Gause's law**, **Coulomb's Law**, **Dipole in a Uniform External Field** are also described.

Here are the important NCERT notes from this chapter-

- **Electric Charges**
- **Maxwell's Equations**
- **Conservation of Charge**
- **Electric dipole**
- **Electric flux**
- **Electric field**
- **Applications of Gauss's Law**
- **Electrostatic Potential and Capacitance**

NCERT Solution: **NCERT Solution Chapter 1 Electric Charges and Fields**

Class 12 Physics Notes for Chapter 2: Electrostatic Potential and Capacitance

In Class 12 Physics Chapter 2, the ideas of **Electrostatic Potential and Capacitance** are discussed. Some of the related concepts are **Van de Graff Generator**, **Potential due to an Electric Dipole**. Important notes for the Electrostatic Potential and Capacitance chapter include-

- **Electrostatic Potential**
- **Equipotential Surfaces**
- **Unit of Capacitance**
- **Energy Stored in a Capacitor**
- **Dielectric Constant**
- **Combination of Capacitors**
- **Electrostatic Shielding**
- **Potential due to a Point Charge**

NCERT Solution: **NCERT Solutions for Class 12 Physics Chapter 2 Electrostatic Potential and Capacitance**

Class 12 Physics Notes for Chapter 3: Current Electricity

The Class 12 Physics Chapter 3 includes the topics **Electric Current**, **Wheatstone Bridge**, **Potentiometer**, **Resistor Colour Codes**. In this chapter, you will also learn concepts like **Ampere**, **Draft Velocity**, and **Electromotive Force**. Some important topics from this chapter are:

- **Kirchhoff's Rules**
- **Ohm's Law And its Limitations**
- **Meter Bridge**
- **Electric Current Formula**
- **Cells, emf, Internal Resistance**
- **Combination of Resistors-Series and Parallel**
- **Cells in Series and in Parallel**
- **Static Electricity**
- **Difference between emf and voltage**

NCERT Solution: **NCERT Solution Class 12 Physics Chapter 3 Current Electricity**

Class 12 Physics Notes for Chapter 4: Moving Charges and Magnetism

In this Class 12 Physics Chapter, the basic concepts of **Magnetism** with the ideas of **Lorentz Force**, **Savart Law**, **Ampere's Circuital Law** are explained. Here are the important Noytes from this chapter-

- **Magnetic Force and Magnetic Field**
- **SI Unit of Magnetic Field**
- **The Moving Coil Galvanometer**
- **The Solenoid and the Toroid**
- **Derivation of lorentz transformation**
- **Magnetic Field on the Axis of a Circular Current Loop**
- **Cyclotron**

NCERT Solution: **NCERT Solution Class 12 Physics Chapter 4 Moving Charges and Magnetism**

Class 12 Physics Notes for Chapter 5: Magnetism and Matter

This Class 12 Physics chapter includes the basic relationships between magnetism and various matters. The related concepts like **Magnetism and Gauss's Law**, **Ferromagnetism**, **Magnetic Susceptibility**, **Magnetic Poles** are also included. The important topics from this chapter include-

- **Magnetic Properties of Materials**
- **Magnetisation and Magnetic Intensity**
- **Difference between electromagnet and permanent magnet**
- **Permanent Magnets and Electromagnets**
- **The Earth's Magnetism**

- Curie weiss law
- Magnetometer
- Einstein Field Equation

NCERT Solution: NCERT Solutions Class 12 Physics Chapter 5 Magnetism and Matter

Class 12 Physics Notes for Chapter 6: Electromagnetic Induction

This Class 12 Physics chapter includes the explanations of the concepts like **Magnetic Flux**, **Faraday's Law of Induction**, **Lenz's Law and Conservation of Energy**. This chapter also includes the detailed studies of **Unit of Magnetic Flux**, **Motional emf**. The important topics from this chapter include-

- Electromagnetism
- Faraday's Law of Electromagnetic Induction
- Eddy Currents
- Types of generators
- Energy Consideration
- Magnetic Flux Formula
- Inductance
- AC Generator
- Electromagnetic Damping

NCERT Solution: NCERT Solutions Class 12 Physics Chapter 6 Electromagnetic Induction

Class 12 Physics Notes for Chapter 7: Alternating Current

In this Class 12 Physics chapter, the core concept related to **Alternating Current** is explained. The topics like **Transformers**, **Power Factor**, **Types of AC Motors** are included in this chapter. The important topics from this chapter are:

- LC Oscillations
- Power in AC Circuit: The Power Factor
- Sharpness Of Resonance
- Difference between resistance and impedance
- AC Voltage Applied to an Inductor
- AC Voltage Applied to a Capacitor
- Difference between a capacitor and an inductor
- Relation between line voltage and phase voltage
- AC Voltage Applied to a Series LCR Circuit
- AC Voltage Applied to a Resistor

NCERT Solution: NCERT Solutions For Class 12 Physics Chapter 7: Alternating Current

Class 12 Physics Notes for Chapter 8: Electromagnetic Waves

This Class 12 Physics chapter can be considered as the introductory chapter to the concept of **Electromagnetic Waves**, along with the ideas of **Maxwell Equations**, **Gamma Radiation**. The other important topics from this chapter-

- **Displacement Current**
- **Electromagnetic Spectrum**
- **Electromagnetic Spectrum**
- **Displacement Current**
- **Impedance of free space**
- **X Ray**
- **Infrared Radiation**
- **Electromagnetic Radiation**
- **Electromagnetic Waves MCQ**
- **Radio Waves**

NCERT Solution: **NCERT Solutions For Class 12 Physics Chapter 8 Electromagnetic Waves**

Class 12 Physics Notes for Chapter 9: Ray Optics and Optical Instruments

This Class 12 Physics chapter deals with the ideas of **Refraction** and Reflection. The important topics related to **Ray Optics and Optical Instruments** are-

- **Reflection of Light by Spherical Mirrors**
- **Reflection on a Plane Mirror**
- **Total Internal Reflection**
- **Wave Particle Duality**
- **Optical Instruments**
- **Refraction through a Prism**
- **Difference between simple and compound microscope**
- **Dispersion by a Prism**
- **Uses of Concave Mirror**
- **Uses of Convex Mirror**
- **Refraction At Spherical Surfaces and By Lenses**

NCERT Solution: **NCERT Solutions for Class 12 Physics Chapter 9 Ray Optics and Optical Instruments**

Class 12 Physics Notes for Chapter 10: Wave Optics

The Physics Class 12 chapter of **Wave Optics** includes the study of the relations between light and waves. This chapter also explains concepts like **Huygens Principle**, **Brewster's Law** and **Coherent Sources**. Some important topics are:

- **Polarisation**
- **Diffraction**

- Difference between diffraction and interference
- Coherent and Incoherent Addition of Waves
- Doppler Effect Formula
- Interference of Light Waves and Young's Experiment
- Raman Scattering
- Brewsters law formula

NCERT Solution: NCERT Solutions Class 12 Physics Chapter 10 Wave Optics

Class 12 Physics Notes for Chapter 11: Dual Nature of Radiation and Matter

This Class 12 Physics chapter introduced the concept of **Dual Nature of Radiation and Matter**. The related concepts, which are explained well, include **Experimental Study of Photoelectric Effect**, **Einstein's Photoelectric Equation**. Here are the important topics from this chapter-

- Photoelectric Effect
- De broglie wavelength formula
- Davisson and Germer Experiment
- Electron Emission
- Compton wavelength
- Particle Nature of Light: The Photon
- Planck Equation
- Quantum Theory Of Light

NCERT Solution: NCERT Solutions Class 12 Physics Chapter 11 Dual Nature of Radiation and Matter

Class 12 Physics Notes for Chapter 12: Atoms

This Class 12 Physics chapter includes an in-depth knowledge of Atoms along with other related concepts like **Alpha Particle Mass**, **Modern Physics**, **Fermi Energy**. Here are the important topics from this chapter-

- Alpha-particle Scattering and Rutherford's Nuclear Model of Atom
- Atomic Spectra
- Spectral Series
- Bohr Model of the Hydrogen Atom
- Ionizing radiation
- Bohr Radius
- Laser
- Avogadro's Number
- Quantum mechanics

NCERT Solution: NCERT Solutions Class 12 Physics Chapter 12 Atoms

Class 12 Physics Notes for Chapter 13: Nuclei

The detailed study of the concept of nuclei is explained in this Class 12 Physics topic. It also includes other important topics like **Beta Decay**, **Nuclear Physics**, and **Readioactive Decay**. Here are some of the important topics from this chapter-

- **Mass-Energy and Nuclear Binding Energy**
- **Nuclear Force**
- **Unit of Radioactivity**
- **Radioactive decay formula**
- **Positron**
- **Nuclear Energy**
- **Nuclear Binding Energy**
- **Mass Energy Equivalence**
- **Neutron Mass**
- **Nuclear reactor based on nuclear fission**

NCERT Solution: **NCERT Solutions for Class 12 Physics Chapter 13 Nuclei**

Class 12 Physics Notes for Chapter 14: Semiconductor Electronics

In this chapter, the basic concepts of Semiconductor Electronics are introduced along with other related concepts like **Transitor**, **Fullwave Rectifier**, and **Half Wave Rectifier**. Some important topics from this chapter are mentioned below-

- **Classification of Metals, Conductors, and Semiconductors**
- **Intrinsic Semiconductor**
- **Extrinsic Semiconductor**
- **p-n Junction**
- **Semiconductor Diode**
- **Light Emitting Diode**
- **Zener Diode**
- **Integrated Circuits**
- **Junction Transistor**
- **Schottky diode**
- **Spintronics**
- **Digital Electronics and Logic Gates**
- **Zener Diode as a Voltage Regulator**
- **Difference between npn and pnp transistor**

NCERT Solution: **NCERT Solutions Chapter 14 Semiconductor Electronics: Materials, Devices and Simple Circuits**